

Validity Analysis: MILU

Target context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Overall risk: HIGH

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology △	1	Serious concern	high
Output Content	2	Significant gaps	medium
Output Form △	1	Serious concern	high
Average	2.2		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is a serious, India-grounded benchmark whose taxonomy and content provide partial coverage of the priority subjects (Polity, History, Economy, Hindi grammar, even UPSC-style Ethics) for North-India Hindi-medium central-exam aspirants, with

empirical confirmation of meaningful regional content (UP, Rajasthan items) in the actual Hindi data. However, for THIS deployment its validity is severely constrained by three structural issues: (1) MILU evaluates only MCQ label accuracy [Q55, Q85] while the deployment requires a Hindi-language explanatory rationale [elicitation Q3] — a fundamental output-ontology and output-form mismatch confirmed by the absence of any rationale field in the dataset; (2) the observed Hindi validation sample is 100% machine-translated [Limitation #1] and contains English-vocabulary items mislabeled as Hindi content [DATASET-D40–D42], inflating construct-irrelevant variance for Hindi-medium students; (3) Current Affairs items are dated 2018–2022 [DATASET-D19–D22], making MILU stale on a heavily-weighted UPSC topic. Strengths in input ontology and partial regional content do not compensate for the output-side gaps. MILU can serve as a partial diagnostic for MCQ knowledge in History/Polity/Economy in Hindi, but cannot validate the deployment's primary product (Hindi rationale quality) at all.

Practical Guidance

What This Benchmark Measures

MILU provides domain- and subject-level MCQ accuracy scores in Hindi that meaningfully probe Indian Polity/Constitution, Indian History (ancient/medieval/modern), Indian Economy, Hindi grammar, and a sampling of UPSC-style ethics scenarios — all priority areas for the deployment. Logical Reasoning and Quantitative Aptitude are partially captured under Science and Business Studies domains. Strongest dimensions are input form (text-only Hindi MCQs match the deployment modality) and input ontology (Indian-exam-grounded taxonomy with confirmed empirical coverage of priority subjects).

Construct Depth

The construct depth is shallow for the deployment's actual task. MILU measures a model's ability to select the correct MCQ option in Hindi, but provides zero evidence on whether the model can produce factually accurate, fluent, or pedagogically appropriate Hindi explanations — which is what the deployed app actually delivers. Within the MCQ scope, depth is also limited: no separation of central-exam vs. state-PSC items, no Current Affairs currency mechanism, mixed translation-induced register risk, and weakest model performance reported precisely in the deployment's priority cultural domains [Q6, Q71].

What Else You Need

(1) Build or commission a Hindi rationale-quality evaluation rubric covering factual accuracy, fluency in Manak Hindi, pedagogical clarity, and code-mixing ceiling — directly addressing the OO/OF gaps [WEB-14 confirms no off-the-shelf option]. (2) Construct a current-affairs evaluation harness with rolling 12-month Indian GK content to address the staleness gap [DATASET-D19–D22]. (3) Filter the MILU Hindi subset to remove specialist Engineering/programming items, English-vocabulary items, and South India-specific items not in central-exam scope [DATASET-D31–D42] before computing deployment-relevant accuracy. (4) Recruit Hindi-medium central-exam coaching experts to spot-validate at least 200 items and to produce a small gold-standard rationale set.